

Position Paper: A Personalized Large Language Model (LLM)-Based Chat Companion for Autistic Children Early Intervention

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ABSTRACT

Autism Spectrum Disorder (ASD) is a lifelong neurodevelopmental disorder characterized by restricted and repetitive sensory-motor behaviors and social communication deficits. Though in the recent decade, a rising interest has been demonstrated among the Human-Computer Interaction (HCI) and broader computing communities in involving multitudes of technologies to support autistic individuals especially children with autism over diverse aspects, the technologies are revealed to often reflect the normative expectations of a neurotypical society and be very one-sided. Thus, autistic children are commonly required to learn the "appropriate" norm of social interaction defined by neurotypical society. In this work, instead of "forcing" autistic children to accommodate technology, we propose a self-advocacy chat companion for autistic children based on the Large Language Model (LLM) as an early intervention tool. In the meantime, we hope it could also serve as a source platform that engages neurotypical users (such as parents) in understanding and learning how to interact and communicate with autistic children.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; **Ubiquitous and mobile computing design and evaluation methods**.

KEYWORDS

Autism, Children, Ubiquitous Computing, Large Language model (LLM), Interaction Design, Personalized Training

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1 INTRODUCTION

Autism Spectrum Disorder (ASD), also known as autism, is a lifelong neurodevelopmental condition typically characterized by restricted and repetitive sensory-motor behaviors and social communication deficits [17, 22] with a median prevalence of 100/10,000 worldwide [36]. Symptoms of autism include impairments in social communication and interaction, sensory irregularities, repetitive behaviors, and varying degrees of intellectual disability [21], while the traits of different autistic individuals could vary from very mild to severe.

A rising interest in involving multitudes of technologies to support individuals with autism over various aspects, such as behavior analysis [20, 33, 34], social interaction [11, 13, 15] and intervention [2, 7, 35], have been shown in the recent decade among the Human-Computer Interaction (HCI) and broader computing communities [29], including some of our works [30–32]. However, technologies are revealed to often reflect the normative expectations of the neurotypical society, which primarily perceives autism as a medical condition while requiring 'correction' via changing their behaviors and traits [29]. Consequently, autistic individuals become the secondary audiences to the externally defined objectives, though they are supposed to be the primary beneficiaries of these technologies, which potentially hinders the actual effectiveness.

In more recent, the emergence of the Large Language Model (LLM) with its robustness has drawn significant attention in diverse areas and has started to be ubiquitously used with promising outcomes [8]. Some pilot explorative works have started to apply the LLM to assist individuals with autism, such as [4, 14, 24], while most are aiming at assisting with the daily lives of adolescents with autism spectrum disorder. Still few works have been focused on utilizing the advantages of LLM in early intervention for autistic children while placing autistic children as primary users using the

self-advocacy approach. In the meantime, it remains largely unexplored on studying the impact of integrating LLM-driven conversational agents (CAs) for autistic individuals [9], which motivates our work.

In this proposed work, instead of "forcing" autistic children to accommodate technology, we propose a self-advocacy chat companion for autistic children based on the Large Language Model (LLM) as an early intervention tool. In the meantime, we hope it could further serve as a source platform that engages neurotypical users (such as parents and caregivers) to better understand and learn how to involve and communicate with autistic children. That is, instead of requiring the children to learn the "appropriate" interaction norms perceived by neurotypical adults, we hope that our proposed LLM-driven conversational agent (CA) could offer ground information to assist allistic individuals to better understand and adapt to the modes of social interaction of autistic children.

2 BACKGROUND

2.1 Autism

Though the cause of autism is recognized as early brain development mutation and neural reorganization [5, 22, 26], the diagnosis of autism still largely relies on behavior analysis measures, such as the Screening Tool for Autism in Toddlers and Young Children (STAT), the Autism Diagnostic Observation Schedule Second Edition (ADOS-2) [23], and the American Psychiatric Association's Diagnostic and Statistical Manual of Mental Disorders (DSM)-5 criteria [3], with the lack of reliable biomarkers.

Historically, autism is often treated as a condition that needs medical attention and correction to adapt to the dominant neurotypical community [16, 29], leading to large portions of research targeting at changing the autistic individual behaviors and traits to cater to the "appropriate" norm of social interaction commonly accepted in neurotypical society while ignoring them as primary users of these technologies. *Neurodiversity* [18, 28] is a more recently accepted understanding of autism which treats autism as one of the neurological variations and encourages self-advocacy and self-empowerment for autistic individuals towards technologies [9, 29]. Considering the broadly diverse traits of autistic children [22], who are also the primary beneficiaries of the technologies, involving autistic children in shaping the technology and offering a capability of personalization in the system to be tailored to their interests and needs becomes essential.

2.2 Large Language Model (LLM) and Conversational Agents (CAs)

To avoid judgmental responses, autistic individuals often tend to seek assistance from trusted caregivers (such as parents and siblings) when facing challenges, especially for autistic children. However, such resources are not always immediately available and raise an extra burden for them. Conversational agents (CAs) have been shown to be supportive of alleviating the situation and reducing social pressure for autism [6], such as Siri [25]. However, the lack of unique and diverse responses hinders its further application to the autistic individual [9].

In recent years, the emergence of the Large Language Model (LLM), such as ChatGPT [1], presents prominent advances in natural language processing (NLP) through offering natural human-like conversation interactions with high accuracy [19]. It has drawn significant attention in diverse areas and started to be ubiquitously used with promising outcomes [8]. A few early explorative works have started considering involving LLM in assisting individuals with autism, while most are aiming at assisting with the daily lives of adolescents with autism spectrum disorder. Hassona and Alqaissi's [14] conducted a comprehensive survey for parents and caregivers over the usage of ChatGPT for information retrieval regarding the oral health of autistic individuals. The outcome is promising, while potential limits of LLM are reported in the study, such as biased content, incorrect information, and nondeterministic responses. Lee et al. [9] evaluated the potential usage of ChatGPT for assisting autistic adults in managing their daily activities through focus group interviews and workshops. Their result reported that ChatGPT could motivate autistic individuals to improve their problem-solving skills. Ashraf et al. [4] initiated a pilot study involving the ChatGPT aiming for behavioral therapy. However, still few works have been focused on utilizing the advantages of LLM in the early intervention for autistic children while placing autistic children as the primary users using the self-advocacy approach. In the meantime, it remains largely unexplored on studying the impact of integrating LLM-driven conversational agents (CAs) for autistic individuals [9], which motivates our work.

2.3 Challenges

Though the LLM-driven CAs have indicated early promising results for autistic individuals, such as [10, 24], challenges still need to be solved before the wider adoption. Firstly, with the open and anonymous nature of LLM data sources, the credibility and safety of responses become more vital when implementing it for early intervention for autistic children and should be properly evaluated [9]. In the meantime, recent work [12] has reported dialog generated by the Large Language Model could reflect harmful societal biases towards marginalized communities, which could hinder its actual usage for autistic individuals. In addition, the generative LLM is trained for and targets neurotypical users, which might not be directly suitable and appropriate for autistic individuals especially for autistic children. For example, interpreting or filling incomplete speech by children is not under the initial design of LLM. A personalized-tailored LLM-driven CA that caters to the unique and diverse needs of autistic children should be implemented by considering corresponding intervention modules (such as Social Communication, Emotional Regulation and Transactional Support, SCERTS [27]). In the meantime, implementing safeguards mechanisms for LLM outputs, such as expert monitoring and feedback loops, becomes essential. Though the study has [6] observed possible intervention among conversation agents for adults on the autism spectrum, how to foster autistic children's intervention while engaging in LLM-driven CAs using a more natural conversation interaction remains unexplored and should be carefully designed.

Furthermore, besides using the LLM-driven conversational agent (CA) as a chat companion to potentially offer early intervention for autistic children in a self-advocacy approach, we hope to further

utilize the system as a ground source to assist the allistic individuals to better understand and adapt to the modes of interaction of the autistic children. Properly synthesizing and extracting the information from the model might be another challenge, as well as privacy concerns.

In addition to text, we should consider combinations of text and picture/video generated contents, including some favorite images/animations (e.g., for children who like terriers, the generative images may include a terrier walking) on exploring the potential benefits of multimodal interactions.

3 CONCLUSION

In this position paper, rather than using the approach of "forcing" autistic children to accommodate technology, we propose a self-advocacy chat companion for autistic children based on the Large Language Model (LLM) as an early intervention tool. In the meantime, we hope it could also serve as a source platform that engages neurotypical users (such as parents and caregivers) to better understand and learn how to interact with autistic children. That is, instead of requiring the children to learn the "appropriate" interaction norms perceived by neurotypical adults, we hope that our proposed LLM-driven conversational agent (CA) could offer ground information to assist the allistic individuals to better understand and adapt to the modes of social interaction of the autistic children. We hope this work could also offer some ideas for future work involving LLM-driven conversational agents (CAs) as natural interaction medium for autistic children while benefiting them through understanding their unique and diverse needs and further encouraging self-advocacy and self-empowerment for autistic individuals towards technologies.

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